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Enhanced energy recovery via separate hydrogen and methane production from two-stage anaerobic digestion of food waste with nanobubble water supplementation

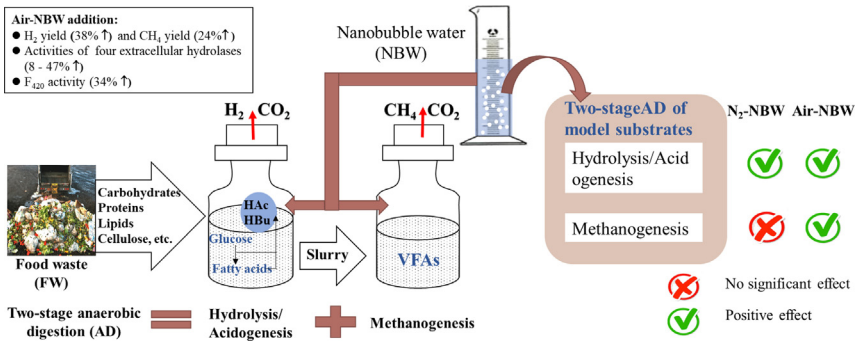
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HIGHLIGHTS

- Air-NBW significantly enhanced H_2 and CH_4 yields from two-stage AD of FW.
- Greater activities of hydrolases were detected under Air-NBW addition.
- Air-NBW positively effected methanogenesis indicated by increased F_{420} activity.

GRAPHICAL ABSTRACT



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ABSTRACT

This study investigated the enhancement effect of N_2- and Air-nanobubble water (NBW) supplementation on two-stage anaerobic digestion (AD) of food waste (FW) for separate production of hydrogen and methane. In the first stage for hydrogen production, the highest cumulative H_2 yield (27.31 ± 1.21 mL/g-VS_{added}) was obtained from FW + Air-NBW, increasing by 38% compared to the control (FW + deionized water (DW)). In the second stage for methane production, the cumulative CH_4 yield followed a descending order of FW + Air-NBW (373.63 ± 3.58 mL/g-VS_{added}) > FW + N_2-NBW (347.63 ± 7.05 mL/g-VS_{added}) > FW + DW (300.93 ± 3.24 mL/g-VS_{added}, control), increasing by 24% in FW + Air-NBW and 16% in FW + N_2-NBW compared to the control, respectively. Further investigations indicate that different gas–NBW may positively impact the different stages of AD process. Addition of N_2-NBW only enhanced the hydrolysis/acidification of FW with no significant effect on methanogenesis. By comparison, addition of Air-NBW promoted both hydrolysis/acidification stage and methanogenesis stage, reflecting by the enhanced activities of four extracellular hydrolases at the end of hydrolysis/acidification and coenzyme F_{420} at the end of methanogenesis, respectively. Results from this work suggest the potential application of Air-NBW in the two-stage AD for efficient renewable energy recovery from FW.
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1. Introduction

Continuous growth of population and the scarcity of resources bring about the exhaustion of energy, thus the importunity and dependence on energy continues to reach record levels. "Turning waste into wealth" opens up a new path for sustainable utilization and management of wastes, which is increasingly attracting the attention of the society nowadays. As known, anaerobic digestion (AD) can convert organic wastes into renewable energy like H_2 and CH_4, which may relieve the pressure of energy scarcity and increase the diversity of fuel sources. From the viewpoint of environmental and economic aspects,

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bioconversion of organic wastes can reduce the risk of environmental pollution and disposal costs (Awasthi et al., 2018).

Municipal solid waste as the main source of organic wastes is estimated to be annually 2.3 billion tonnes by 2025, with food waste (FW) accounting for the largest proportion (≥50%) (Ahamed et al., 2016; Qin et al., 2019). According to the UN Food and Agriculture Organization (FAO), the roughly same quantities of food were dissipated in industrialized and less industrialized countries (670 and 630 million tonnes per annum). However, FW is more problematic in industrialized countries, i.e. those countries with per capita FW by consumers between 95 and 115 kg per annum in Europe and North America as against 6 and 11 kg per annum in the low-income Sub-Saharan Africa and South/Southeast Asia regions (FAO, 2014). If not handled properly, one million tonnes of FW would cause 4.5 million tonnes of CO_2 emission from landfills. In contrast, if being treated by AD, one million tonnes of FW could produce enough electricity to power 29,818 households (Kosseva, 2009). As a result, AD is considered as an appropriate option for the disposal of FW. However, the rapid production of large amounts of volatile fatty acids (VFAs) due to the highly biodegradable components in FW may result in a reduced pH and thus inhibit the activity of methanogens at the early stage of AD, one of the main reasons for the failure of FW-AD reactors (Liu and Liao, 2019). Two-stage AD can separate the complicated AD process into two main stages (hydrolysis/acidification and methanogenesis) to ensure the optimal growth environment for the corresponding microbial groups. In the first stage, the complex organic compounds can be maximally degraded into short-chain VFAs under slightly acidic condition (pH = 5.0–6.0), from which hydrogen as clean energy with high specific energy content (142 kJ/g) can be produced as another important intermediate in addition to VFAs (Silva et al., 2018; Yuan et al., 2019); during the second stage, the produced VFAs from the first stage can be converted into methane by methanogens under relatively neutral condition (pH = 7.0–8.0) (Voelklein et al., 2016). Thus, two-stage AD is deemed to be promising to reduce organic loads and improve overall energy conversion efficiency besides the generation of the two high-calorie gases, i.e. H_2 and CH_4. As reported by Baldi et al. (2019), the biogas production and volatile solid (VS) degradation rate from the two-stage AD of FW and activated sludge were increased by 26% and 9% in comparison to the single-stage AD, respectively.

However, the composition of FW is complex, rich in carbohydrates (41–62%), proteins (15–25%), and lipids (13–30%) (Boonpiyo et al., 2018). Although the high content of carbohydrates in FW can be easily converted, the high contents of proteins, lipids, and some recalcitrant cellulosic materials may greatly reduce the hydrolysis efficiency (Kim et al., 2019). On the other hand, the low hydrogen yield in the first stage might be the major limiting factor for the two-stage AD (Yuan et al., 2019). To enhance hydrolysis and biogas production from AD of FW, various pretreatment methods have been attempted to treat FW. For instance, Ding et al. (2017) examined the effect of hydrothermal pretreatment on FW-AD, attaining the maximum hydrogen and methane yields about 43.0 and 511.6 mL/g-VS under the optimum condition (140 °C, 20 min), which increased by 24% and 32% when compared with the untreated FW, respectively. Acidic (pH = 2) and alkaline (pH = 11–12) pretreatments are also frequently adopted to FW due to their simple operations (Kim et al., 2014; Jang et al., 2015), from which hydrogen production can be largely improved while adversely impacted by operation mode. Although these pretreatments improved the hydrolysis of FW and thus enhanced biogas production to some extent, the difficulty in maintaining the operation conditions and the addition of expensive chemicals greatly hinder their practical application.

Recently, nanobubble water (NBW), i.e. the water containing bubbles with diameter ranging from several to hundreds of nm, has been applied for environmental remediation due to its unique characteristics (such as the long-term stable existence of NBs and the generation of hydroxyl radicals) and zero pollution (Azevedo et al., 2019; Hu and Xia, 2018; Li et al., 2014). As known, organic solid wastes with high solid content usually require a large volume of water/wastewater addition prior to AD due to mass transfer limitations (Bollon et al., 2013). Using NBW instead of other water or wastewater can improve the transfer of intermediates to bacterial sites because of its high mass transfer ability (Temesgen et al., 2017). It is worth mentioning that addition of NBW has also been proven as an environmentally friendly method to enhance methane yield from AD of organic solid wastes such as waste activated sludge (WAS) and cellulosic materials (Wang et al., 2019; Yang et al., 2019; Wang et al., 2020a; Wang et al., 2020b). These previous works proved that supplementation of various gas-NBW can improve the hydrolysis of organic solids with enhanced methane production. For instance, Yang et al. (2019) found that addition of N_2-NBW improved the activities of extracellular hydrolases and VFAs production during AD of WAS. Wang et al. (2020a) reported that addition of Air-NBW enhanced the cellulose crystallinity reduction by 20% and the hydrolytic and methanogenic bacteria were enriched in the high cellulose loading AD system. Up to the present, however, no report is available about the effect of NBW on hydrogen production from AD, especially the two-stage AD process of high organic content wastes (like FW in this study).

In this study, N_2-NBW and Air-NBW were selected for the tests due to their different initial anaerobic (N_2-NBW) or micro-aerobic environment (Air-NBW) created after addition. Both hydrogen and methane production from the two-stage AD of FW were monitored. The underlying mechanisms of NBW addition to each stage were further explored for the first time by using the model substrates including protein (bovine serum albumin, BSA), glucose, and acetic acid (HAc) in the separate hydrolysis/acidogenesis and methanogenesis experiments. Finally, the enzyme activities with or without NBW addition were also evaluated to evidence the above mechanisms.

2. Materials and methods

2.1. FW, inoculum and NBW

FW was directly collected from the dormitory kitchen at the University of Tsukuba, Japan, and the anaerobically digested sludge as inoculum was sampled from a wastewater treatment plant in Tsuchiura, Ibaraki Prefecture, Japan. The details of preparation and storage of FW and inoculum have been described previously (Hou et al., 2020). The physicochemical characteristics of FW and inoculum are shown in Table 1.

The NBW, i.e. N_2-NBW or Air-NBW was prepared using the NBW generator (HACK FB11, Japan) under the optimal generation conditions with NBs size of ~140 nm and the bubble particle concentration >100 million particles/mL, following the specific manufacturing process according to previous works (Wang et al., 2019; Yang et al., 2019). Air used for the Air-NBW generation was taken directly from the natural environment of the laboratory, and the purity of N_2 used for the N_2-NBW generation was ≥99.9995%.

2.2. Experimental set-up of the two-stage AD of FW system

2.2.1. Batch biochemical hydrogen potential (BHP) tests

The Schott Duran serum bottles with identical specifications (250 mL in total volume with 150 mL of working volume) were used

Table 1
Physicochemical characteristics of the food waste and inoculum used in this study.

Parameter	Food waste	Inoculum
Total solids (TS, %)[a]	13.08 ± 0.12	0.70 ± 0.01
Volatile solid (VS, %)[a]	8.93 ± 0.18	0.45 ± 0.03
Total volatile fatty acids (tVFAs, mg/L)	5287.74 ± 67.57	9.77 ± 1.70
Soluble total organic carbon (sTOC, mg/L)	10191.74 ± 21.63	171.61 ± 0.22

[a] Wet weight basis.

in both stages of experiments. In the BHP tests, each bottle was loaded with the same amount (20.0 g wet weight) of FW and 60 mL of inoculum. After that, 70 mL of deionized water (DW), N_2-NBW, or Air-NBW was added to make the total working volume of each bottle to be 150 mL, in which the DW group was used as a blank control. In addition, the initial total VS in each bottle was 13.96 ± 0.49 g/L and the feed-to-microorganisms (F/M) ratio was about 6.5 (VS basis). As reported by Yuan et al. (2019), a high F/M ratio could inhibit the activity of methanogens in the untreated inoculum, and achieve a higher hydrogen yield. More specifically, all the initial pH values were adjusted to 5.50 ± 0.10 using a 2 M HCl solution in the BHP tests, which is regarded as the optimal pH for the enzyme activity of hydrogenase and also could inhibit the activity of methanogens to avoid methane production (Liu et al., 2019; Micolucci et al., 2014).

2.2.2. Batch biochemical methane potential (BMP) tests

In the BMP tests, the remaining acidified slurries or mixtures from the BHP tests were used as substrates for the second stage of AD to produce methane. 80 mL of acidified slurry and 50 mL of inoculum (at an F/M ratio (VS basis) of 4.5 and initial total VS of 7.18 ± 0.40 g/L) were added into each bottle and then 20 mL of DW, N_2-NBW, or Air-NBW was added to make the total working volume of each bottle to be 150 mL. In this stage, all the initial pH values were adjusted to 7.50 ± 0.10 using a 2 M NaOH or HCl solution, which has been reported as the optimal pH for methanogens (Liu et al., 2019; Micolucci et al., 2014). Additionally, the inoculum was activated for about two weeks under mesophilic condition (38 ± 1 °C) until no gas production, and the total solid (TS) and VS of inoculum were about 0.67% ± 0.01% and 0.42% ± 0.01% after activation, respectively.

Before starting the experiments, all the bottles were flushed with N_2 gas (purity ≥99.9995%) for 2 min for 3 times to ensure anaerobic conditions and then incubated in a temperature-controlled incubator (38 ± 1 °C) for 68 h during the first-stage of AD for hydrogen production and 24 d during the second-stage of AD for methane production. In the BHP tests, the first gas sample was taken 8 h later, and then regularly collected over time every 12 h to detect the contents of H_2 and CO_2. In the BMP tests, the gas sample was regularly collected over time every 2 d to detect the contents of CH_4 and CO_2.

2.3. Effect of NBW addition on each stage in the two-stage AD by using the model substrates

The two-stage AD consists of two steps: hydrolysis/acidification and methanogenesis. The hydrolysis stage that can break down carbohydrates, proteins, and lipids into monosaccharides, amino acids, and long-chain fatty acids is crucial for complex organic compounds like FW (Dong et al., 2019). In this experiment, the effect of NBW addition on each stage in the two-stage AD system was investigated by using the model substrates (protein (BSA), glucose, and HAc).

In this part, all the experiments were carried out in 100 mL glass serum bottles, and divided into three groups (Group-1, Group-2, and Group-3). Group-1 was carried out to illustrate the effect of NBW addition on the hydrolysis process. In this group, 1500 mg/L of protein as the model substrate together with 25 mL of inoculum was loaded. Moreover, the total working volume was made up to 50 mL with 25 mL of DW, N_2-NBW, or Air-NBW, respectively. The concentration of protein in the filtrated supernatant from each bottle was daily monitored during the 8 days' hydrolysis.

Group-2 was conducted to evaluate the effect of NBW addition on the acidogenesis step and glucose (1500 mg/L) was used as the model substrate. All the other operation conditions were the same as the hydrolysis step. During the examination, the concentration of glucose in the filtrated supernatant from each bottle was analyzed by sampling every 3 h for 12 h.

Group-3 was used to assess the effect of NBW addition on methanogenesis. HAc was used as the model substrate in this group because of the fact that HAc was the main contributor to methane production according to the VFAs measurement in this study (Section 3.1.2). A mixture with 90 mg of HAc (1500 mg/L) and 30 mL of inoculum was added into the designated bottle, then 30 mL of DW, N_2-NBW, or Air-NBW was added to make the total volume up to 60 mL. The biogas composition was analyzed every 2 days to calculate the cumulative methane yield during the 10 days' methanogenesis stage.

Before starting all the above experiments, each bottle was firstly sealed with a rubber stopper and resin crimp cap, and then flushed with high-purity N_2 gas (≥99.9995%) for 1 min for 3 times to remove oxygen and finally placed in a temperature-controlled incubator (38 ± 1 °C).

2.4. Microbial enzyme activities at each stage

To a great extent, the stability and efficiency of AD of organic wastes depend on the activity of microorganisms, and the main functional microorganisms in each stage are different in the two-stage AD process (Ma et al., 2019). Thus, the major microbial enzyme activities were evaluated in this study, aiming to provide insights into the effect of NBW addition on the two-stage AD of FW.

Four extracellular hydrolases (at the end of the first stage), i.e. alkaline phosphatase (ALP), acid phosphatase (ACP), α-glucosidase, and protease, and coenzyme F_{420} (at the end of the second stage) were measured. For their analysis, the activities of ALP, ACP, and α-glucosidase were measured according to Goel et al. (1998). 0.1% *p*-nitrophenyl phosphate disodium salt (pnpP), 0.1% pnpP, and 0.1% *p*-nitrophenyl-α-D-glucopyranoside were used as substrates for the determination of ALP, ACP, and α-glucosidase, respectively. Besides, the activity of protease was analyzed according to the Folin-Phenol reagent method with 10 mg/mL of casein as the substrate (Dai et al., 2017; Ledoux and Lamy, 1986). Finally, the activity of coenzyme F_{420} was measured according to the method described by Delafontaine et al. (1979) and Dong et al. (2019).

In this study, the relative enzyme activity was calculated to evaluate the effect of NBW addition on microorganisms according to Eq. (1).

$$\text{Relative enzyme activity}\ (\%) = \frac{A_{NBW}}{A_{DW}} \times 100 \quad (1)$$

where A_{NBW} and A_{DW} represent the absorbance at the same wavelength of the NBW added sample and DW added sample, respectively.

2.5. Other analytical methods and calculations

The TS and VS of FW and inoculum were measured based on the standard methods (APHA, 2012). All liquid samples were filtered through 0.45 μm membrane filters before being analyzed and all the measurements were performed in triplicate. The concentrations of protein and glucose were determined by Lowry's method and Phenol-sulfuric acid method, respectively (Dubois et al., 1956; Lowry et al., 1951). Soluble total organic carbon (sTOC) was determined by a total carbon analyzer (TOC-VCSN with ASI-V autosampler, Shimadzu, Japan). pH value was determined with a semi-solid pH meter (Testo 206, Germany). According to the detailed description by Huang et al. (2016), the concentration of individual VFA, namely HAc, propionic acid (HPr), iso-butyric acid (*iso*-HBu), *n*-butyric acid (*n*-HBu), iso-valeric acid (*iso*-HVa), or *n*-valeric acid (*n*-HVa), was determined by a gas chromatograph (GC-8A, Shimadzu, Japan) equipped with Unisole F-200 30/60 column and flame ionization detector.

For the gas samples, the volume of biogas production was measured using 50 and 20 mL of gas-tight syringes and normalized to standard temperature and pressure (25 °C, 1 atm). The contents of H_2, CH_4, and CO_2 in biogas were analyzed using a gas chromatograph (GC-8A, Shimadzu, Japan) after sampling 1 mL biogas produced, and the volume

of hydrogen or methane was calculated according to Eq. (2) (Fu et al., 2020).

$$V_t = C_t \cdot V_{biogas,t} + V_{head} \cdot (C_t - C_{t-1}) \quad (2)$$

Additionally, the production of hydrogen or methane during the interval between *t* and *t-1* was calculated based on VS_{added} in this study, according to Eq. (3).

$$Y_t = V_t / \text{Total } VS_{added} \quad (3)$$

where V_t (H_2 or CH_4) (mL) is the volume of hydrogen or methane produced during the interval between times *t* and *t-1*; C_t (%) and C_{t-1} (%) are the contents of hydrogen or methane at times *t* and *t-1*; $V_{biogas,\ t}$ (mL) is the volume of biogas produced at time *t*; V_{head} (mL) is the headspace volume of the bottle; Y_t (H_2 or CH_4, mL/g-VS_{added}) is the yield of hydrogen or methane during the interval between times *t* and *t-1* based on the total amount of VS_{added}.

In the BHP and BMP tests, the experimental data of cumulative hydrogen or methane yield in each stage was fitted to the modified Gompertz model as in Eq. (4) (Xing et al., 2017).

$$H_t \text{or } M_t = P \ exp\left\{ - exp\left[\frac{R_{max}}{P}(\lambda - t)\right]\right\} \quad (4)$$

where H_t or M_t (mL/g-VS_{added}) is the specific hydrogen or methane yield at a given time *t*; *P* (mL/g-VS_{added}) is the maximum hydrogen or methane production potential; R_{max} (mL/(g-VS_{added}·h) or mL/(g-VS_{added}·d)) is the maximum hydrogen or methane production rate; λ (h or d) is the lag time; *t* (h or d) is the experimental duration; and *e* (Euler's number) = 2.7183.

In this study, the energy recovery from the two-stage AD of FW with or without the NBW addition was calculated according to Eq. (5).

$$E_r = 12.78 \times Y_{H_{2,yield}} + 40.03 \times Y_{CH_{4,yield}} \quad (5)$$

where E_r (kJ/g-VS_{added}) is the energy recovery from the two-stage AD of FW; $Y_{H2,\ yield}$ (L/g-VS_{added}) is the hydrogen yield in the first stage; and $Y_{CH4,\ yield}$ (L/g-VS_{added}) is the methane yield in the second stage. The heating value of H_2 (12.78 kJ/L) or CH_4 (40.03 kJ/L) under 1 atm and 25 °C was estimated according to the density and calorific value of H_2 or CH_4 gas, i.e. 0.09 or 0.72 kg/m^3 and 142 or 55.6 kJ/g, respectively (Silva et al., 2018).

2.6. Statistical analysis

All the experiments were carried out in duplicate and the samples were measured in triplicate. The data were processed by Microsoft Excel 2010 and expressed as the mean value ± standard deviation (SD). One-way analysis of variance (ANOVA) was also applied for statistical analysis and statistical significance was assumed at $p < 0.05$. Origin 9.0 (Origin Lab, trial version) was used to draw and analyze the fitting curves.

3. Results and discussion

3.1. Hydrogen production from the first stage of two-stage AD of FW

3.1.1. Hydrogen yield and kinetics analysis

The results of hourly H_2 production and cumulative H_2 yield in the BHP tests with addition of DW, N_2-NBW, or Air-NBW are shown in Fig. 1. It's important to mention that no methane production was detected during the whole H_2 production process. The hourly H_2 productions from all the bottles were almost identical during the first 20 h ($p = 0.98 > 0.05$) and then increased to the peak values of 1.09 ± 0.00, 1.26 ± 0.04, and 1.34 ± 0.00 mL/(g-VS_{added}·h) at the 32nd h in FW + DW, FW + N_2-NBW, and FW + Air-NBW, respectively (Fig. 1a). In addition, hourly H_2 productions from the NBW added reactors were also higher than that from the control with DW addition after reaching the peak values. These results indicate that addition of NBW can enhance H_2 production. Similarly, the cumulative H_2 yields from the NBW added reactors were greater than that from the DW added reactor (Fig. 1b). It is interesting to notice that the maximum cumulative H_2 yield (27.31 ± 1.21 mL/g-VS_{added}) was obtained in the Air-NBW added reactor, about 38% higher than that from the control (19.79 ± 0.07 mL/g-VS_{added}). Meanwhile, the cumulative H_2 yield (25.42 ± 0.37 mL/g-VS_{added}) from the N_2-NBW added reactor was 28% higher than that from the control. Some researchers reported that OH• radicals were produced via collapse of cavity bubbles, which can be explained by the Young-Laplace equation (Ghadimkhani et al., 2016). Therefore, the enhancement of cumulative H_2 yield by NBW addition may be due to the fact that NBs in the NBW will break up when contacted with organic matters in FW, resulting in the generation of OH• free radicals that can help oxidize and decompose the organic matters, leading to accelerated hydrolysis of FW (Ghadimkhani et al., 2016; Takahashi et al., 2007). On the other hand, the uneven mass transfer of organic matters in FW is regarded as one of the main reasons for its low biogas production

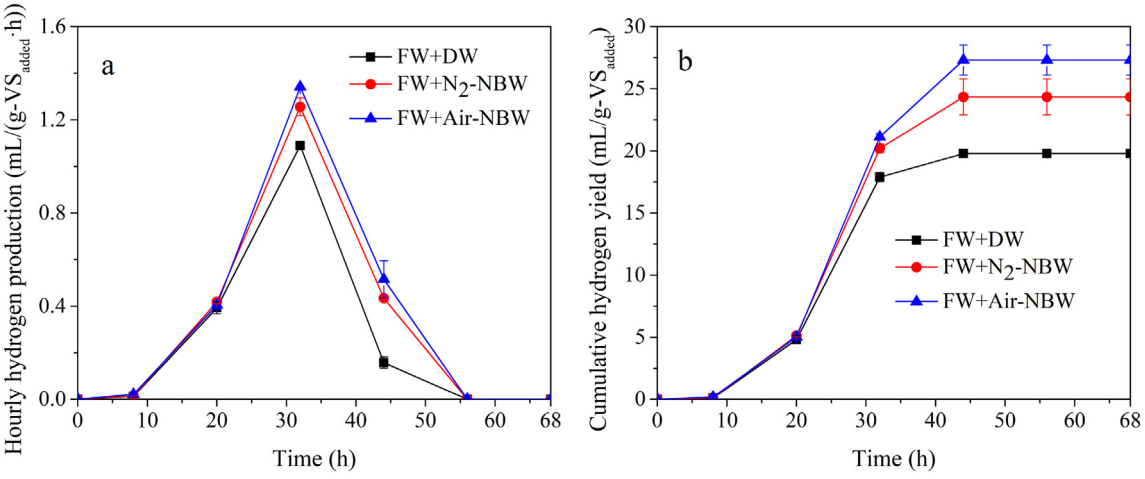


Fig. 1. Time courses of hourly H_2 production (a) and cumulative H_2 yield (b) under DW and NBW (N_2-NBW and Air-NBW) addition. DW, deionized water; FW, food waste; NBW, nanobubble water.

efficiency. A previous work by Hu and Xia (2018) indicated that micro-nanobubbles might promote mass transfer efficiency, achieving enhanced utilization and conversion efficiency of organic substrates. In comparison, although N_2-NBW can ensure a strictly anaerobic environment, the cumulative H_2 yield was 10% lower than that from the Air-NBW added reactor. This implies that the introduction of trace amounts of air (or O_2) would increase rather than inhibit hydrogen production. This observation is consistent with Fu et al. (2020) who obtained the maximum hydrogen yield of 24.3 mL/g VS at an oxygen loading of 5 mL/g-VS, about 70% higher than that without micro-aeration during the two-stage AD of corn straw. As reported, improved H_2 production under the micro-oxygen condition can be attributable to the rapid hydrolysis of carbohydrates and proteins (Sarkar and Venkata Mohan, 2017). Besides, Johansen and Bakke (2006) and Rafieenia et al. (2017) claimed that proper aeration can enhance the activities of hydrolytic enzymes like cellulase and protease. In this study, the related results are shown in Sections 3.3 and 3.4. Restated, the micro-aerobic environment was realized by adding Air-NBW in these tests, which needs further in-depth investigations before being applied in practice.

The relevant parameters of the first stage were obtained by fitting the BHP experimental data to the Modified Gompertz model (Table 2). The results show that the highest P_{H2} (27.66 mL/g-VS_{added}) and $R_{H2,\ max}$ value (4.37 mL/(g-VS_{added}·h)) were obtained in the Air-NBW added reactor, which was consistent with the actual experimental results in the BHP tests. In addition, the lag time (λ) was reduced slightly in the NBW added reactors (16.88–17.10 h) when compared to the control (17.56 h). It indicates that although the addition of NBW resulted in some change of the original environment for the microorganisms, it didn't affect the start-up of the reactor.

3.1.2. VFAs evolution in the first-stage AD

The individual VFA and total VFAs concentration at the end of BHP tests are shown in Fig. 2a. In comparison to the control, a higher total VFAs concentration was detected under NBW addition, increasing by about 9% and 10% in FW + N_2-NBW and FW + Air-NBW, respectively. It is worth noting that HAc and HBu were the major VFAs during the hydrogen production stage, amounting to >90% of the total VFAs concentration, in agreement with previous observations (Luo et al., 2010; Yuan et al., 2019).

In this study, the highest HAc production (2954.09 ± 48.34 mg/L) was detected in FW + Air-NBW, which is corresponding to its highest H_2 production. However, the HBu concentration followed a descending order of FW + N_2-NBW (1889.62 ± 0.45 mg/L) > FW + Air-NBW (1775.84 ± 61.51 mg/L) > FW + DW (1583.59 ± 2.12 mg/L). This means that addition of Air-NBW is more conducive to HAc production, while addition of N_2-NBW is beneficial for HBu production. The molar ratio of HBu to HAc (B/A ratio), as a proportional indicator of H_2 yield from per mole hexose (Ghimire et al., 2015), is plotted in Fig. 2b. Only a slight increase in B/A ratio was noticed in the NBW added reactors, which was 0.40, 0.45, and 0.42 in FW + DW, FW + N_2-NBW, and FW + Air-NBW, respectively. This result indicates that addition of NBW may not alter the metabolic pathway to produce hydrogen, and the increase of hydrogen yield might be contributed by the improved hydrolysis as discussed later.

Table 2
Parameters estimated by fitting the experimental data from the first-stage and second-stage AD tests to the modified Gompertz model.

Stage/reactors	Parameters in the modified Gompertz model			
The first-stage AD of FW				
	P_{H2} (mL/g-VS_{added})	$R_{H2,\ max}$ (mL/(g-VS_{added}·h))	λ (h)	R^2
FW + DW	19.84 ± 0.05	3.87 ± 0.08	17.56 ± 0.08	0.9999
FW + N_2-NBW	24.52 ± 0.15	4.32 ± 0.14	16.88 ± 0.22	0.9996
FW + Air-NBW	27.66 ± 0.29	4.37 ± 0.22	17.10 ± 0.41	0.9990
Stage/reactors	Parameters in the modified Gompertz model			
The second-stage AD of FW				
	P_{CH4} (mL/g-VS_{added})	$R_{CH4,\ max}$ (mL/(g-VS_{added}·d))	λ (d)	R^2
FW + DW	300.93 ± 2.78	100.29 ± 4.10	4.02 ± 0.17	0.9981
FW + N_2-NBW	346.58 ± 3.49	133.90 ± 6.77	4.05 ± 0.19	0.9974
FW + Air-NBW	375.25 ± 3.01	174.84 ± 8.33	4.44 ± 0.15	0.9981

3.2. Methane production from the second-stage of the two-stage AD of FW

3.2.1. Daily methane production and cumulative methane yield

The remaining hydrolyzed and acidified organic slurries or mixtures at the end of BHP tests were further used as the substrates for methane production in the subsequent BMP tests. The results of daily CH_4 production and cumulative CH_4 yield are presented in Fig. 3. Regarding the daily CH_4 production (Fig. 3a), all the test reactors behaved almost same during the first 6 days and reached their peak values on day 8. However, FW + Air-NBW achieved the highest peak value of 133.90 ± 0.56 mL/(g-VS_{added}·d), in comparison to 87.22 ± 6.28 and 115.17 ± 0.20 mL/(g-VS_{added}·d) from FW + DW and FW + N_2-NBW, respectively. This result is in agreement with the VFAs concentrations in the three reactors (Fig. 2a). It has been claimed that a high VFAs concentration is beneficial for methane production (De Gioannis et al., 2017). The BMP tests lasted 24 d and the cumulative CH_4 yields from the NBW added reactors were significantly higher than the control (with DW added, $p = 0.01 < 0.05$, Fig. 3b). Similarly, the cumulative CH_4 yield followed a descending order of FW + Air-NBW (373.63 ± 3.58 mL/g-VS_{added}) > FW + N_2-NBW (347.63 ± 7.05 mL/g-VS_{added}) > FW + DW (300.93 ± 3.24 mL/g-VS_{added}), increasing by 24% and 16% when compared with FW + DW, respectively. This observation clearly shows that addition of NBW can positively effect both hydrogen and methane productions in the two-stage AD of FW, especially under Air-NBW addition. This result also proves once again that the micro-aerobic environment created by Air-NBW addition can promote rather than inhibit AD process. Similar phenomenon could be found in a previous study (Wang et al., 2020b), in which the enhancement of methane production from AD of cellulose was achieved with the supplementation of O_2-containing gas NBW. At the end of methanogenesis, the final VS degradation rates in FW + DW, FW + N_2-NBW, and FW + Air-NBW were 65.97% ± 0.92%, 71.18% ± 0.04%, and 74.33% ± 0.21%, respectively, indicating that addition of NBW can improve VS decomposition thus enhance hydrogen and methane productions from the two-stage AD of FW. The above results also suggest that NBW-based two-stage AD process can successfully improve energy recovery from the AD of FW with no chemicals addition and no secondary pollution.

3.2.2. Kinetics involved in biomethane production

The methanogenesis stage was also described with the parameters of P, R_{max}, and λ by fitting the experimental data to the Modified Gompertz model as shown in Table 2. The highest P_{CH4} (375.25 mL/g-VS_{added}) and $R_{CH4,\ max}$ (174.84 mL/(g-VS_{added}·d)) were also obtained in the Air-NBW added reactor, followed by the N_2-NBW added reactor (P_{CH4} = 346.58 mL/g-VS_{added}, $R_{CH4,\ max}$ = 133.90 mL/(g-VS_{added}·d)) and the control (P_{CH4} = 300.93 mL/g-VS_{added}, $R_{CH4,\ max}$ = 100.29 mL/(g-VS_{added}·d)). When compared to the control, a remarkable increase in $R_{CH4,\ max}$ by about 34% was obtained in the N_2-NBW added reactor, and an increase by about 74% was achieved in the Air-NBW added reactor. This observation further confirms that addition of NBW is beneficial for methane production. In addition, the lag phase of methanogenesis was not significantly different among the three test conditions, about 4.02–4.44 d.

3.3. Effect of NBW addition on the specific individual step involved in the two-stage AD by using the model substrates

In this study, the effects of NBW addition on the hydrolysis/acidogenesis and methanogenesis were explored by using the model

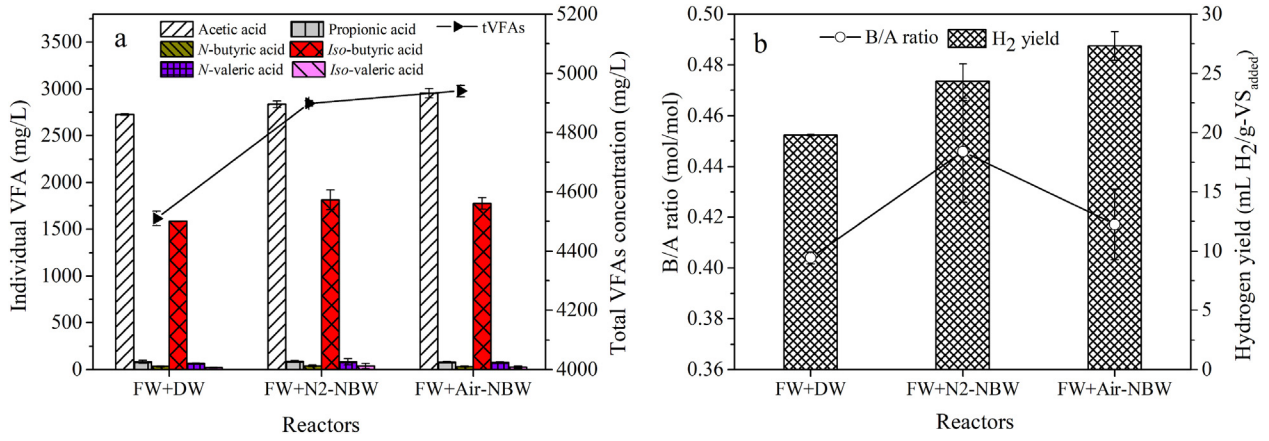


Fig. 2. The individual VFA and total VFAs concentrations (a) and the molar HBu/HAc (B/A) ratio and H_2 yield (b) at the end of the first stage under DW and NBW (N_2-NBW and Air-NBW) addition. FW, food waste; VFA, volatile fatty acid; DW, deionized water; NBW, nanobubble water.

substrates and the results are shown in Fig. 4. As seen, the hydrolysis and acidogenesis steps were significantly enhanced with addition of NBW, especially under Air-NBW addition ($p = 0.02$ and $0.01 < 0.05$). The concentration of protein was detected to decrease by 1190.18 ± 36.07, 1397.56 ± 45.08, and 1433.62 ± 9.02 mg/L, achieving protein degradation rates of 67%, 79%, and 83% in FW + DW, FW + N_2-NBW, and FW + Air-NBW, respectively (Fig. 4a). Similarly, the concentration of glucose decreased by 1098.54 ± 76.5, 1208.7 ± 39.78, and 1315.8 ± 0.00 mg/L, resulting in glucose consumption rates of 69%, 76%, and 83% in FW + DW, FW + N_2-NBW, and FW + Air-NBW, respectively (Fig. 4b). The increase in degradation and consumption rates of these two model substrates further illustrates that addition of NBW enhanced the hydrolysis and acidogenesis during the two-stage AD process. For the methanogenesis (Fig. 4c), the cumulative methane yields were 94.89 ± 4.51, 100.93 ± 4.92, and 126.79 ± 5.12 mL/g-HAc in FW + DW, FW + N_2-NBW, and FW + Air-NBW, respectively. Most recently, Yang et al. (2019) found that addition of N_2-NBW did not significantly enhance the methanogenesis of acetate, which agreed with the result from this study. The cumulative methane yield was only increased by 6% in the N_2-NBW added reactor when compared to the control ($p = 0.85 > 0.05$) in this study. However, the methanogenesis was significantly enhanced under Air-NBW addition as its cumulative methane yield increased by 33% ($p = 0.04 < 0.05$). Therefore, the positive effect of N_2-NBW addition on the two-stage AD of FW could attribute to the enhancement on hydrolysis/acidogenesis, while addition of Air-NBW can not only promote the hydrolysis/acidogenesis of FW but also enhance the methanogenesis. This result was further confirmed by the enzyme activity as shown in Section 3.4.

3.4. Effects of NBW addition on the microbial enzyme activities at each stage

The relative activities of ALP, ACP, α-glucosidase, and protease at the end of the first stage (hydrolysis/acidification) and coenzyme F_{420} at the end of the second stage (methanogenesis) were determined and calculated to provide insights into the effects of NBW on microorganisms in the two-stage AD of FW. Clearly, the activities of four extracellular hydrolases under NBW addition were higher than those in the control (Fig. 5a). In particular, addition of Air-NBW achieved the highest activities of ALP, ACP, α-glucosidase, and protease, increasing by 29%, 47%, 14%, and 8% in comparison to the control, respectively. It has been reported that phosphatases including ALP and ACP can catalyze the hydrolysis of organic phosphate esters, and protease is capable of hydrolyzing soluble proteins into amino acids (Mu and Chen, 2011; Wan et al., 2020). In addition, the activity of α-glucosidase in FW + N_2-NBW increased by 23% compared with the control. α-Glucosidase is considered as a crucial enzyme for the conversion of disaccharides and oligosaccharides into monosaccharides (Ni et al.,

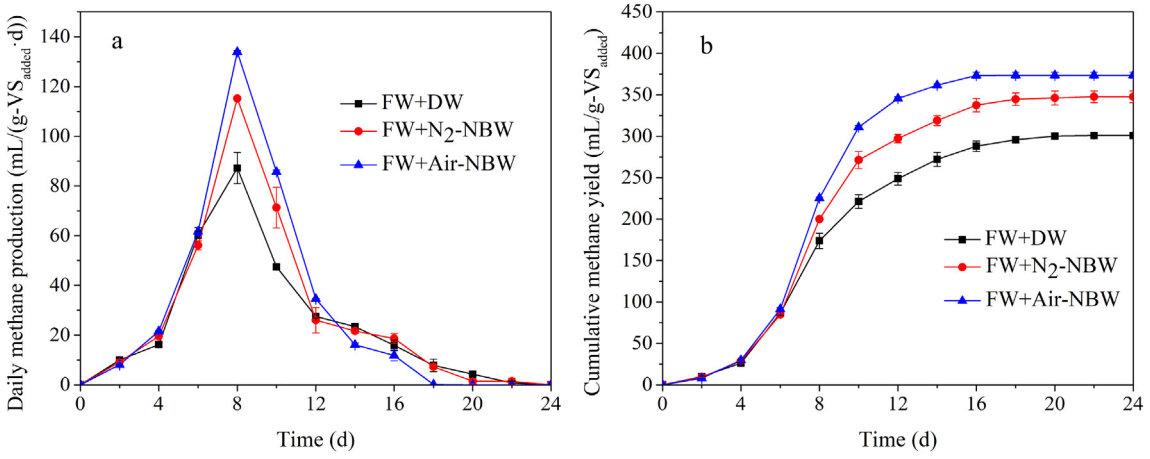


Fig. 3. Time courses of daily CH_4 production (a) and cumulative CH_4 yield (b) under DW and NBW (N_2-NBW and Air-NBW) addition. DW, deionized water; FW, food waste; NBW, nanobubble water.

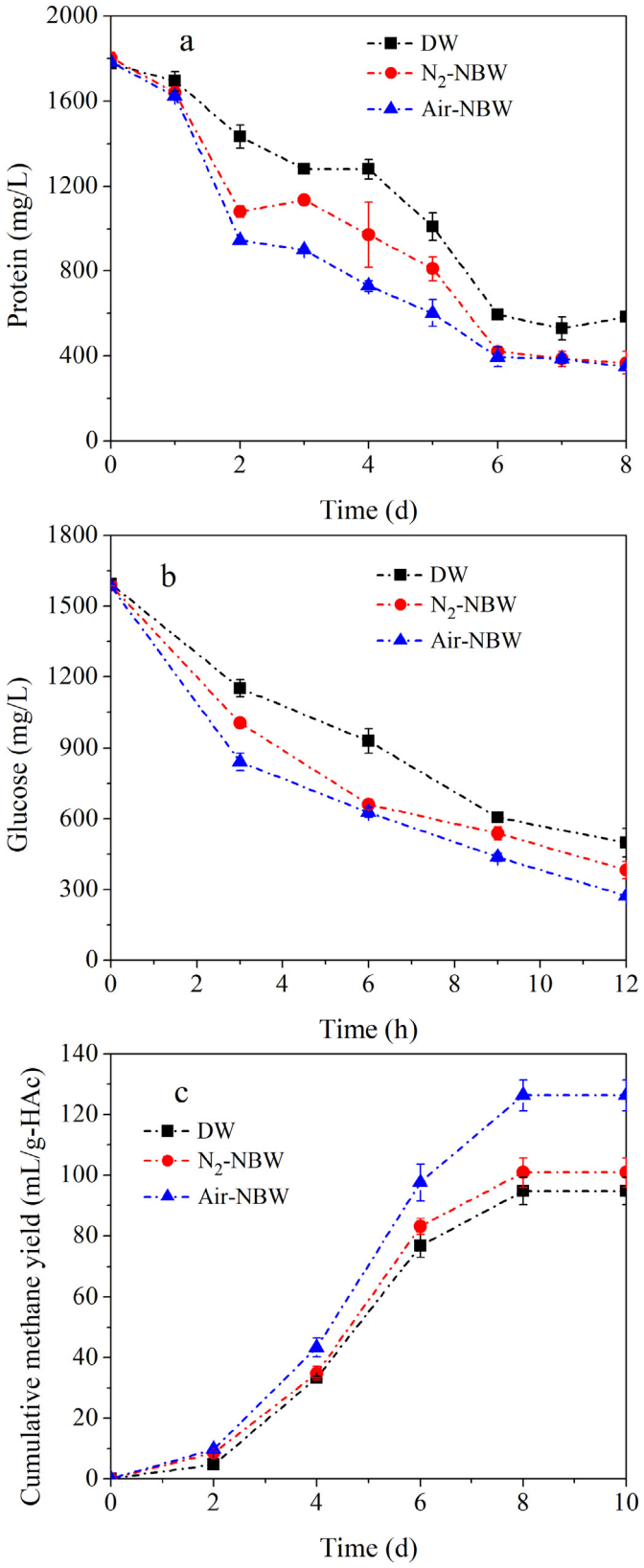


Fig. 4. Effects of NBW (N_2-NBW and Air-NBW) addition on the specific individual step involved in the two-stage anaerobic digestion by using the model substrates: (a) variation of protein concentration during hydrolysis step; (b) variation of glucose concentration during acidogenesis step; and (c) cumulative methane yield from HAc during methanogenesis. DW, deionized water; NBW, nanobubble water.

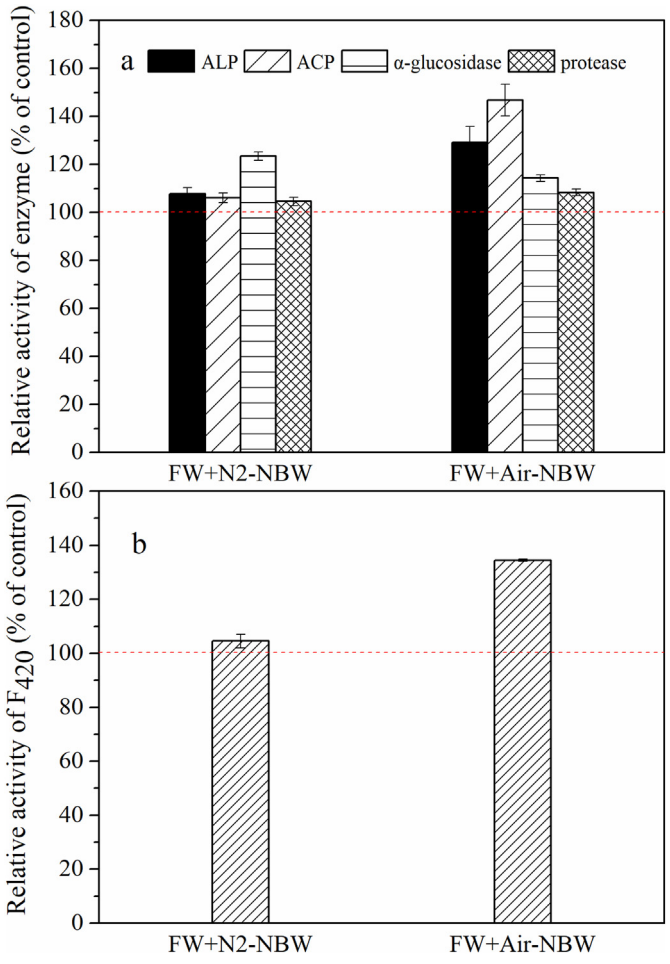


Fig. 5. Effects of NBW (N_2-NBW and Air-NBW) addition on the relative activities of ALP, ACP, protease, α-glucosidase, and protease at the end of the first stage (hydrolysis/acidification, a), and the relative activity of coenzyme F_{420} at the end of the second stage (methanogenesis, b). FW, food waste; NBW, nanobubble water; ALP, alkaline phosphatase; ACP, alkaline phosphatase.

2020). The increase of bioactivity in these hydrolases verifies that addition of NBW could enhance the hydrolysis of FW under the test first-stage conditions.

As it is known, coenzyme F_{420} plays a critical role in the process of methanogenesis, which represents the conversion activity of the electron donor-driven by redox proton translocation in the methanogenic Archaea (Mu and Chen, 2011). Methanogens have been generally considered as strictly anaerobic microbes and trace amounts of oxygen may be toxic to them (Botheju, 2011). However, Fig. 5b shows that the relative activity of coenzyme F_{420} increased by 4% and 34% in FW + N_2-NBW and FW + Air-NBW in comparison to the control, respectively. As reported, *Methanobacterium* and *Methanosarcina* can tolerate limited oxygen environment (Botheju, 2011; Fu et al., 2016). Due to addition of Air-NBW in the first stage, the microorganisms may have chance to expose to the micro-oxygen environment; thus aerobic or facultative anaerobic bacteria present in the first stage can grow fast and consume the limited oxygen quickly, resulting in quick transformation of the micro-oxygen environment into anaerobic condition in the second stage. In this study, addition of Air-NBW had a significantly positive effect on the activity of coenzyme F_{420} with improved methane production. This observation further confirms that the improvement of hydrogen and methane productions from the two-stage AD of FW through Air-NBW supplementation may involve in the enhancement of microbial activities in both stages, namely hydrolysis/acidification and methanogenesis.

Table 3
Energy recovery from the two-stage AD of FW with or without NBW addition.

Reactors	Energy recovery (kJ/g-VS_{added})		
	H_2 production in the first stage	CH_4 production in the second stage	Two-stage AD
FW + DW	0.25 ± 0.00	12.05 ± 0.13	12.30 ± 0.13
FW + N_2-NBW	0.32 ± 0.00	13.92 ± 0.28	14.24 ± 0.28

As seen from the above results, Air-NBW is more promising than N_2-NBW for the two-stage AD of FW to produce hydrogen and methane sequentially due to the following two reasons. Firstly, the air is preferable and free everywhere, which is easier to obtain than nitrogen. Thus, the manufacturing of Air-NBW can reduce cost and operation difficulty compared with that of N_2-NBW. Secondly, addition of Air-NBW showed superior effectiveness than that of N_2-NBW in the two-stage AD of FW tests either in terms of hydrogen production or methane production.

3.5. Energy recovery from the two-stage AD of FW with or without NBW addition

The energy recovery from H_2 and CH_4 in the two-stage AD of FW with or without NBW addition was analyzed as shown in Table 3. It is worth noting that the two-stage AD of FW achieved much higher energy recovery (12.30 kJ/g-VS_{added} in FW + DW) than one-stage AD using the same FW (9.68 kJ/g-VS_{added}) (Hou et al., 2020). This observation to some extent is in agreement with a previous work by Yuan et al. (2019) who reported 10% higher energy yield obtained in the two-stage AD of FW than the one-stage AD of FW. Similarly, De Gioannis et al. (2017) adopted two-stage AD of FW and obtained 20% increase in energy recovery when compared with the one-stage AD of FW. As pointed out by Fu et al. (2020), the first stage of two-stage AD can be regarded as biological pretreatment of high organic content wastes (such as FW in this study), in which hydrogen energy recovery and substrates pretreatment are realized simultaneously. In this study, owing to addition of Air-NBW, the increment in energy recovery from the two-stage AD of FW is about 58% than the one-stage AD of FW. This indicates that addition of Air-NBW can greatly improve the performance of two-stage AD of FW, achieving more energy recovery from FW when compared with one-stage AD. In this study, the highest total energy recovery was obtained from FW + Air-NBW (15.31 ± 0.16 kJ/g-VS_{added}), increasing by 24% when compared to FW + DW. Previous studies have reported that technologies such as hydrothermal, acidic/alkaline pretreatment, and aeration can improve the energy yields from anaerobic digestion of FW. Rafieenia et al. (2017) claimed that the pre-aerated (at an air flow rate of 5 L/h for 24 h) protein rich FW was the best in terms of total energy generation, amounting to 9.64 kJ/g-VS. Ding et al. (2017) applied hydrothermal pretreatment (140 °C for 20 min) on FW, and gained 18.8 kJ/g-VS energy yield. In fact, supplementation of Air-NBW can also be considered as an environmentally friendly technology due to no chemicals addition and no secondary pollution with more renewable energy recovery from the two-stage AD of FW. Provided that the increased energy output outweighs the additional energy demand used for manufacturing NBW, addition of NBW in industrial application has practical significance. Therefore, the economic assessment of NBW addition in the two-stage AD of FW for co-production of hydrogen and methane was conducted in this study, and Air-NBW addition was used as an example.

The production capacity of the NBW generator is 12–22 L/min (the median generation capacity (17 L/min) was used for calculation) at a power consumption of 0.75 kW. When 1 t of FW is treated, the required 4500 L of Air-NBW will consume 3.31 kWh of electricity. The comparative economic assessment results between FW + DW and FW + Air-NBW are shown in Table 4. The generated electricity from FW + Air-NBW increased about 23% when compared to FW + DW.

4. Conclusions

Addition of NBW can achieve higher H_2 and CH_4 yields from the two-stage AD of FW than the control, especially under Air-NBW addition which increased H_2 yield by 38% and CH_4 yield by 24%. The relative activities of hydrolases increased by 8–47% in the first stage, and the activity of coenzyme F_{420} was significantly enhanced by 34% in the second stage with addition of Air-NBW. Therefore, the developed two-stage AD of FW with Air-NBW supplementation can enhance hydrogen and methane production simultaneously, greatly increasing energy recovery (by 23%) from the whole AD system. This two-stage AD process is a promising and environmentally friendly approach, targeting the maximization of renewable energy recovery from FW.

CRediT authorship contribution statement

Tingting Hou: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing. **Jiamin Zhao:** Methodology, Data curation. **Zhongfang Lei:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration. **Kazuya Shimizu:** Writing - review & editing, Supervision. **Zhenya Zhang:** Writing - review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 4
Comparative results of economic assessment between FW + DW and FW + Air-NBW when treating 1 t of FW.

Reactors	H_2 yield (L)	Heating value of H_2 (kJ)	CH_4 yield (L)	Heating value of CH_4 (kJ)	Total heating value (kJ)	Electricity generation (kWh)
FW + DW	1767	22,582	26,873	1,075,726	1,098,308	305.09
FW + Air-NBW	2439	31,170	33,365	1,335,601	1,366,771	376.35

The unit conversion from kJ to kWh is 1/3600.

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